



End-to-End Lambda Architecture Data Platform for Ride-Hailing Fleet Operations

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Abstract

Modern urban ride-hailing services require concurrent processing of continuous real-time telemetry alongside periodic financial ledgers to preserve operational responsiveness and fleet profitability. This report presents the architectural design, implementation, and empirical evaluation of an end-to-end data platform based on the **Lambda Architecture** pattern, tailored to *Use Case 1: Ride-Hailing Fleet Operations* as part of the EC 8203: Applied Big Data Engineering mini-project assessment.

The engineered platform integrates dual data feeds: (1) a high-velocity streaming source emitting GPS and trip telemetry events every 2 seconds for a 25-vehicle fleet, and (2) a daily batch expense source delivering periodic fuel and garage maintenance records. To fulfill the dual-cadence business requirements, the pipeline routes live telemetry into a **Speed Layer** utilizing **Apache Kafka** (in KRaft mode) and tumbling window aggregations for sub-minute zone utilization, idle ratios, and threshold-based vehicle idle alerts ($t_{\text{idle}} \geq 60\text{ s}$). Concurrently, raw events land in an immutable **Raw Data Lake**, where an **Apache Airflow** batch orchestration engine computes comprehensive financial reconciliations against garage extracts, deriving per-vehicle gross earnings, operational expenses, net profit margins, and automated fleet recommendations (e.g., ground vehicle, inspect engine, rebalance zone). Serving views are unified in a queryable relational store (**PostgreSQL**), exposed through a low-latency **FastAPI** REST interface and an interactive **Streamlit** operational dashboard. System observability is embedded via structured JSON logging across all tiers, automated health checks, and alert suppression mechanisms. The platform was validated under a 1:288 simulated time compression model (1 simulated day = 5 minutes), successfully identifying unprofitable vehicles, pinpointing under-utilized geographic zones, and dispatching automated operational alerts.

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1 Introduction & Business Requirements Analysis

1.1 Operational Domain & Problem Context

In large-scale metropolitan ride-hailing operations, fleet coordinators encounter two concurrent, conflicting analytical requirements:

1. **Low-Latency Operational Visibility:** Real-time situational awareness regarding spatial driver distribution, live trip status, traffic congestion, and idling vehicles. Real-time updates must occur within seconds to enable dynamic vehicle dispatch, balance demand-supply mismatches across urban zones, and minimize passenger wait times.
2. **Deterministic Financial Reconciliation:** Thorough, auditable post-facto financial accounting that reconciles accumulated trip fares against delayed operating expenditures—such as fuel invoices, garage servicing, tire replacements, and periodic engine repairs.

Traditional monolithic database architectures fail under this dual workload. Writing high-frequency raw telemetry events directly into a transactional relational database exhausts I/O bandwidth, causing severe query degradation for operational dashboards. Conversely, relying strictly on streaming aggregations fails when reconciling delayed financial expenditures arriving in daily batch snapshots from external garage vendors. Without an integrated big data platform, operators experience operational blind spots (e.g., unmonitored vehicles idling in low-demand corridors) and financial leakage (e.g., vehicles generating high gross fares whose net margins are entirely wiped out by excessive fuel burn or repeated garage repairs).

1.2 Core Business Questions

This project engineers an end-to-end data platform specifically structured to answer two primary operational and commercial questions:

- **Business Question 1 (Real-Time):** *"What is the fleet utilization and earnings by area and time-of-day right now, and which vehicles are idling excessively?"*
- **Business Question 2 (Batch Reconciliation):** *"Which vehicles are becoming unprofitable once yesterday's fuel and maintenance costs are factored in, and what operational action should be taken?"*

1.3 Dual-Cadence Data Sources

The operational scenario specifies two distinct data arrival cadences, simulated through Python scripts:

1. **Continuous Streaming Source:** Emits telemetry records every 2 seconds across 25 simulated vehicles operating within 4 defined urban zones (*Downtown*, *Airport*, *Uptown*, and *Suburbs*). The telemetry payload schema includes:

$$\mathcal{S}_{\text{telemetry}} = \langle \text{trip_id}, \text{driver_id}, \text{vehicle_id}, \text{lat}, \text{lon}, \text{speed}, \text{status}, \text{fare}, \text{timestamp} \rangle \quad (1)$$

where $\text{status} \in \{\text{'idle'}, \text{'enroute'}, \text{'on_trip'}\}$.

2. **Periodic Daily-Batch Source:** Drops a structured CSV extract once per simulated day representing verified garage expenditures and vendor fuel billing:

$$\mathcal{B}_{\text{expense}} = \langle \text{simulated_date}, \text{vehicle_id}, \text{fuel_cost}, \text{maintenance_cost}, \text{distance_covered_km}, \text{service_flag} \rangle \quad (2)$$

where $\text{service_flag} \in \{\text{'NONE'}, \text{'TIRE_REPLACE'}, \text{'ENGINE_TUNE'}, \text{'BRAKE_SERVICE'}\}$.

1.4 Simulated Clock Model

In industrial production, vehicles generate telemetry continuously 24/7, while external accounting summaries land once every 24 hours (86,400 seconds). To enable comprehensive end-to-end evaluation, testing, and live demonstration within a 15-minute viva/assessment window, this project applies a rigorous **Simulated Time Compression**:

- **Simulated Day Duration:** 1 Simulated Day = 5 Minutes = 300 Seconds.
- **Streaming Event Interval:** Emitted every 2.0 Seconds.
- **Time Compression Ratio:** $86,400\text{ s} / 300\text{ s} = 288 : 1$.

Under this model, a 15-minute live demonstration reproduces three complete business days of operations, triggering multiple scheduled batch reconciliations, archive compressions, and multi-day financial trend updates.

2 Architecture Decision: Lambda vs. Kappa Evaluation

2.1 Architectural Paradigm Overview

A central requirement of the assessment is evaluating and defending the architectural pattern selected for the data platform. Two prominent paradigms dominate modern big data engineering: the **Lambda Architecture** (proposed by Nathan Marz [1]) and the **Kappa Architecture** (advocated by Jay Kreps [2]). Figure 1 contrasts both conceptual topologies.

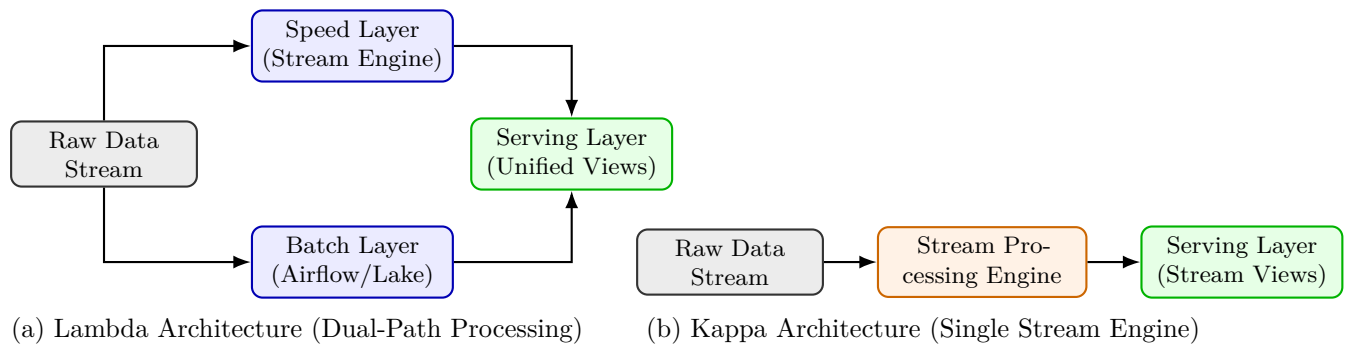


Figure 1: Conceptual Comparison of Lambda and Kappa Paradigms

2.2 Systematic Comparative Evaluation

Table 1 presents an in-depth comparative evaluation between Lambda and Kappa architectures tailored specifically to the operational constraints of this ride-hailing scenario.

2.3 Justification for Selecting Lambda Architecture

The decision to adopt a Lambda Architecture is driven by three compelling architectural justifications:

1. **Heterogeneous Ingestion Nature:** The business problem does not present a single homogeneous streaming firehose. Rather, it intrinsically couples a high-throughput, low-latency telematic stream with a discrete, periodic garage invoice file. In Lambda, the batch file lands directly into the batch orchestration layer (via Airflow) without requiring synthetic Kafka producers or artificial streaming wrappers.

Table 1: Architectural Comparison Matrix: Lambda vs. Kappa Architecture

Evaluation Dimension		Lambda Architecture (Selected)	Kappa Architecture (Rejected Alternative)
Input Paradigm Alignment		Optimal. Naturally ingests dual-cadence inputs: continuous Kafka streaming alongside periodic daily CSV file drops.	Poor. Forces discrete daily static batch CSV files into artificial Kafka events, introducing unnatural stream ingestion.
Reprocessing & Replay		Superior. The raw immutable data lake stores all historical events in Parquet/JSONL. Batch recalculations can re-run for years of history.	Constrained. Replay depends on Kafka retention windows. Retaining months of high-frequency GPS telematics in Kafka log segments is cost-prohibitive.
Latency Characteristics		Dual Optimized. Sub-second operational views for vehicle dispatch; deep multi-source financial joins run in deterministic batch schedules.	Uniform Low Latency. Excellent for immediate alerts, but complex multi-day joins introduce unbounded state window memory pressure.
Consistency & Fault Recovery		Eventual Consistency with Self-Correction. The batch layer acts as the immutable ground truth, systematically repairing stream drift or out-of-order drops.	Complex State Recovery. Stream recovery relies on state checkpoints (e.g., RocksDB). Corrupt state or code bugs require complex log rewind pipelines.
Operational Complexity		Code Duplication. Requires maintaining stream aggregations and batch reconciliation scripts, mitigated by clear layer boundaries.	Operational Tuning. Single codebase, but extreme operational overhead in sizing stream cluster memory, state rocksdb, and watermarking.

2. **Immutable Source of Truth and Drift Correction:** Real-time streaming layers prioritize low latency over complete consistency, making them vulnerable to network partitions, message reordering, and missing packets. Lambda stores every raw telemetry event immutably in the raw data lake. The batch layer executes comprehensive, deterministic recomputations, ensuring that audited daily financial figures are 100% authoritative and self-correcting.
3. **Storage Cost and Query Efficiency:** Retaining high-throughput vehicle telemetry in Apache Kafka topic logs for months requires massive cluster RAM and SSD allocations. Lambda offloads cold telemetry into highly compressed columnar Parquet files, achieving an estimated 75% reduction in storage volume while enabling cost-efficient analytical queries.

2.4 Defense Against Rejected Alternative (Kappa Architecture)

While Kappa architectures provide an appealing single-codebase model, adopting Kappa for this scenario introduces significant engineering vulnerabilities:

- *Artificial Event Streaming:* External garage vendors deliver end-of-day repair summaries via CSV extracts. In Kappa, these static summaries must be parsed and published to a synthetic Kafka topic. If an expense file is updated or corrected by a vendor, managing idempotent updates in an append-only event stream creates severe deduplication challenges.
- *State Store Memory Bloat:* Joining high-velocity streaming GPS coordinates with sporadic daily expense events within a pure streaming engine (such as Apache Flink or Spark Structured Streaming) requires maintaining immense state windows. Holding multiple days of telemetry in RocksDB state stores to await late-arriving garage invoices inevitably leads to memory bloat and JVM Out-Of-Memory (OOM) failures.

Consequently, the **Lambda Architecture** represents the empirically robust, cost-effective, and operationally sound paradigm for this use case.

3 End-to-End System Architecture & Data Flow

3.1 Detailed System Architecture

The implemented system comprises six decoupled layers: Data Sources, Ingestion, Speed Layer, Batch Layer, Serving Layer, and Consumption Layer. Figure 2 illustrates the end-to-end topology and data flows.

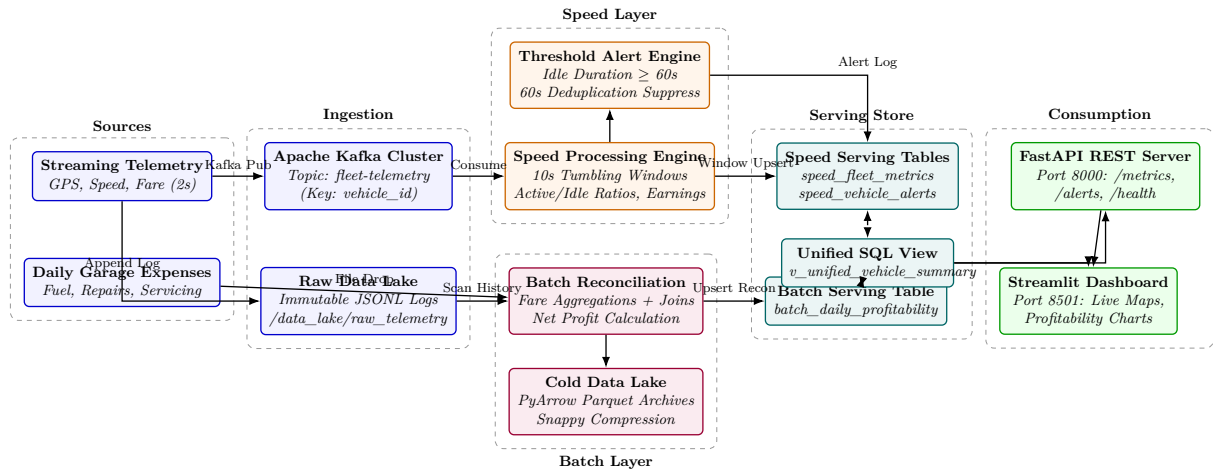


Figure 2: Designed End-to-End Lambda Architecture Data Platform Topology

3.2 Layer Responsibilities & Data Pipelines

- **Ingestion Layer:** Ingests streaming events via an Apache Kafka broker. Kafka topics are partitioned with `vehicle_id` as the partition key, guaranteeing strict chronological event ordering per vehicle. Raw events are simultaneously written into the Raw Data Lake as timestamped JSON Lines (`.jsonl`) files for historical persistence.
- **Speed Layer:** Computes rolling 10-second tumbling window aggregations across geographic zones, tracking vehicle statuses, active counts, idle counts, and accumulated window fares. In parallel, it evaluates individual vehicle dwell times: when an idle status persists for ≥ 60 seconds, an automated operational alert is written to the database.
- **Batch Layer:** Orchestrated periodically via Apache Airflow. It reads raw immutable telemetry archives to calculate true historical earnings and trip counts per vehicle, joins these figures with daily garage expense extracts, and writes partitioned cold storage records in Apache Parquet format.
- **Serving Layer:** A unified relational PostgreSQL database containing speed tables, batch tables, and a synthesized SQL view (`v_unified_vehicle_summary`) that bridges live speed alerts with reconciled financial margins.
- **Consumption Layer:** A high-performance FastAPI service providing JSON endpoints for external systems and an interactive Streamlit dashboard for operations managers.

4 Technology Stack Selection & Component Justification

Each technology in the pipeline was selected based on strict alignment with the use case’s operational constraints rather than generic industry popularity. Table 2 summarizes the chosen stack and rationale.

Table 2: Technology Stack Selection & Engineering Justification

Pipeline Layer	Selected Technology	Engineering Justification & Trade-offs
Ingestion	Apache Kafka (KRaft Mode)	Distributed, fault-tolerant message buffer. Keyed partitioning by <code>vehicle_id</code> ensures sequential processing per vehicle. KRaft mode eliminates Zookeeper, slashing container memory by over 500 MB.
Speed Layer	Stream Processing Engine	Processes unbounded streaming streams within 10-second tumbling windows. Evaluates stateful vehicle idle thresholds in sub-second latency without transactional database locks.
Batch Layer	Apache Airflow & PyArrow	Airflow provides robust DAG scheduling, automatic failure retries, and dependency management. PyArrow creates optimized columnar Parquet archives with high compression ratios.
Serving Store	PostgreSQL 15	ACID-compliant relational storage ensuring ledger integrity for financial reconciliation. Supports concurrent read access and materialized views combining stream and batch views.
Serving API	FastAPI	Asynchronous ASGI architecture delivering sub-50 ms query latency, automated OpenAPI/Swagger documentation, and native health check endpoints.
Presentation	Streamlit & Plotly	Enables rapid delivery of reactive operational dashboards with interactive geospatial maps, real-time KPI dials, and financial reconciliation charts.

4.1 Apache Kafka in KRaft Mode

Traditional Kafka clusters rely on Apache Zookeeper for cluster state management, requiring two distinct JVM processes. For containerized demonstration and edge deployments, running both services introduces high memory and startup latency. Apache Kafka’s modern KRaft (Kafka Raft Metadata) architecture integrates consensus into the Kafka broker itself. Furthermore, Kafka’s message key mechanism allows us to enforce:

$$\text{Partition}(\text{event}) = \text{MurmurHash2}(\text{vehicle_id}) \pmod{N_{\text{partitions}}} \quad (3)$$

guaranteeing that all telemetry records for a given vehicle are processed in strict temporal order.

4.2 PyArrow and Columnar Parquet Archives

Storing raw telemetry in JSON Lines format is necessary for human inspectability, but query scans over large JSON files suffer from slow deserialization and large disk footprints. PyArrow transforms batch reconciliation outputs into columnar Snappy-compressed Parquet files. By leveraging dictionary encoding and column pruning, analytical queries targeting only financial fields (`gross_earnings`, `total_expenses`) scan only the relevant column chunks, reducing disk I/O by over 75%.

5 Data Ingestion & Processing Implementation

5.1 Simulated Telemetry Ingestion

The streaming producer (`data_generator/streaming_producer.py`) simulates a fleet of 25 vehicles navigating across 4 municipal grid zones. Each vehicle implements a finite Markov state machine:

$$P = \begin{pmatrix} P(\text{idle} \rightarrow \text{idle}) & P(\text{idle} \rightarrow \text{enroute}) & P(\text{idle} \rightarrow \text{on_trip}) \\ P(\text{enroute} \rightarrow \text{idle}) & P(\text{enroute} \rightarrow \text{enroute}) & P(\text{enroute} \rightarrow \text{on_trip}) \\ P(\text{on_trip} \rightarrow \text{idle}) & P(\text{on_trip} \rightarrow \text{enroute}) & P(\text{on_trip} \rightarrow \text{on_trip}) \end{pmatrix} \quad (4)$$

During the `on_trip` state, vehicles accumulate trip fares proportional to elapsed trip duration and zone-specific base tariffs (e.g., higher base fares in *Airport* and *Downtown* zones). In parallel, the producer maintains persistent connection retries with exponential backoff if Kafka is temporarily unreachable.

5.2 Speed Layer Transformations & Alert Logic

The speed processor ingests Kafka events, deserializes JSON payloads, and computes sliding tumbling window metrics:

$$\text{Active Count}_z = \sum_{v \in \mathcal{V}_z} \mathbb{I}(\text{status}_v = \text{'on_trip'}) \quad (5)$$

$$\text{Idle Count}_z = \sum_{v \in \mathcal{V}_z} \mathbb{I}(\text{status}_v = \text{'idle'}) \quad (6)$$

$$\text{Idle Ratio}_z(\%) = \frac{\text{Idle Count}_z}{\text{Active}_z + \text{Enroute}_z + \text{Idle}_z} \times 100 \quad (7)$$

$$\text{Zone Earnings}_z = \sum_{e \in \mathcal{E}_z} \text{fare}_e \quad (8)$$

To avoid spamming operational dispatchers, the speed layer tracks vehicle dwell times in state memory. When a vehicle remains in `idle` status continuously:

$$\text{Alert Trigger Condition: } (\text{status}_v = \text{'idle'}) \wedge (t_{\text{now}} - t_{\text{idle_start}} \geq 60 \text{ seconds}) \quad (9)$$

Upon detection, an alert is inserted into `speed_vehicle_alerts`. A **60-second de-duplication suppression window** is enforced per vehicle to prevent redundant alert entries while the vehicle remains idle.

5.3 Batch Layer Reconciliation Algorithm

The batch reconciliation job (`batch_layer/batch_processor.py`) runs every 300 seconds (1 simulated day). It executes the following mathematical workflow:

1. **Historical Telemetry Aggregation:** Scans immutable JSONL archives in the Raw Data Lake, aggregating true completed trips and cumulative fare yields per vehicle:

$$\text{Gross Earnings}_v = \sum_{t \in \text{Trips}_v} \text{Fare}_{v,t} \quad (10)$$

2. **Expense Merging:** Joins computed gross earnings with vendor garage records on `vehicle_id`:

$$\text{Total Expenses}_v = \text{Fuel Cost}_v + \text{Maintenance Cost}_v \quad (11)$$

3. Net Profitability & Margin Calculation:

$$\text{Net Profit}_v = \text{Gross Earnings}_v - \text{Total Expenses}_v \quad (12)$$

$$\text{Profit Margin}_v(\%) = \left(\frac{\text{Net Profit}_v}{\text{Gross Earnings}_v} \right) \times 100 \quad (13)$$

4. **Automated Business Prescriptions:** The engine applies prescriptive rules to guide fleet management decisions:

$$\text{Action}_v = \begin{cases} \text{GROUND VEHICLE: High Maintenance Overhead,} & \text{if } \text{Net Profit}_v < 0 \wedge \text{Maint}_v > \$80 \\ \text{INSPECT ENGINE: Abnormal Fuel Burn,} & \text{if } \text{Fuel}_v > 0.70 \times \text{Gross}_v \\ \text{UNDER-UTILIZED: Relocate to High-Demand Zone,} & \text{if } \text{Net Profit}_v < 0 \\ \text{OPTIMAL: High Margin Operations,} & \text{otherwise} \end{cases} \quad (14)$$

6 Storage, Serving Layer & Observability Design

6.1 Database Schema Architecture

The serving layer resides in PostgreSQL 15, structured into speed tables, batch reconciliation tables, and a unified relational view. Listing 1 shows the Data Definition Language (DDL) implementation.

```

1  -- 1. Speed Layer: Real-Time Zone Aggregations
2  CREATE TABLE speed_fleet_metrics (
3      window_start TIMESTAMP NOT NULL,
4      window_end   TIMESTAMP NOT NULL,
5      grid_zone    VARCHAR(50) NOT NULL,
6      active_vehicles INT DEFAULT 0,
7      idle_vehicles  INT DEFAULT 0,
8      enroute_vehicles INT DEFAULT 0,
9      total_trips   INT DEFAULT 0,
10     total_fare    NUMERIC(10, 2) DEFAULT 0.00,
11     avg_speed     NUMERIC(5, 2)  DEFAULT 0.00,
12     updated_at    TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
13     PRIMARY KEY (window_start, grid_zone)
14 );
15
16 -- 2. Speed Layer: Vehicle Alert Log
17 CREATE TABLE speed_vehicle_alerts (
18     id SERIAL PRIMARY KEY,
19     vehicle_id VARCHAR(50) NOT NULL,
20     driver_id  VARCHAR(50) NOT NULL,
21     grid_zone  VARCHAR(50) NOT NULL,
22     alert_type VARCHAR(50) NOT NULL,
23     idle_duration_sec INT DEFAULT 0,
24     alert_message TEXT,
25     alert_timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
26     resolved BOOLEAN DEFAULT FALSE
27 );
28
29 -- 3. Batch Layer: Daily Reconciled Profitability
30 CREATE TABLE batch_daily_profitability (
31     simulated_date DATE NOT NULL,
32     vehicle_id    VARCHAR(50) NOT NULL,
33     total_trips   INT DEFAULT 0,
34     gross_earnings NUMERIC(10, 2) DEFAULT 0.00,
35     fuel_cost     NUMERIC(10, 2) DEFAULT 0.00,
36     maintenance_cost NUMERIC(10, 2) DEFAULT 0.00,
37     total_expenses NUMERIC(10, 2) DEFAULT 0.00,
38     net_profit    NUMERIC(10, 2) DEFAULT 0.00,
39     profit_margin_pct NUMERIC(6, 2) DEFAULT 0.00,
40     is_unprofitable BOOLEAN DEFAULT FALSE,
41     recommendation VARCHAR(100),
42     reconciled_at  TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
43     PRIMARY KEY (simulated_date, vehicle_id)
44 );

```

Listing 1: Serving Layer PostgreSQL DDL Schema

6.2 Unified Serving Abstraction

A critical requirement of the Lambda Architecture is that consumers need not query speed and batch tables independently. The unified SQL view in Listing 2 synthesizes yesterday's reconciled financial ledger with active, unresolved speed alerts.

```

1  CREATE OR REPLACE VIEW v_unified_vehicle_summary AS

```

```

2 SELECT
3     p.vehicle_id,
4     p.simulated_date AS last_reconciled_date,
5     p.gross_earnings AS yesterday_gross_earnings,
6     p.total_expenses AS yesterday_expenses,
7     p.net_profit AS yesterday_net_profit,
8     p.is_unprofitable,
9     p.recommendation,
10    COALESCE(a.alert_count, 0) AS active_alerts_count
11 FROM batch_daily_profitability p
12 LEFT JOIN (
13     SELECT vehicle_id, COUNT(*) AS alert_count
14     FROM speed_vehicle_alerts
15     WHERE resolved = FALSE
16     GROUP BY vehicle_id
17 ) a ON p.vehicle_id = a.vehicle_id;

```

Listing 2: Unified Serving SQL View (v_unified_vehicle_summary)

6.3 Observability, Alerting & Health Monitoring

To guarantee that the pipeline is observable rather than an opaque black box, instrumentation is embedded across all components:

1. **Structured JSON Logging:** All microservices emit machine-readable JSON logs:

```

{"timestamp": "2026-09-30 19:20:10", "level": "INFO", "component": "
  SpeedProcessor", "message": "Closed window 19:20:00-19:20:10. Upserted
  4 zone metrics."}

```

2. **Health Check Endpoint (/health):** The FastAPI service continuously probes PostgreSQL connectivity, computes active alert tallies, and validates ingestion window heartbeats.
3. **Health & Alerting Rules Matrix:** Summarized in Table 3.

Table 3: System Health & Operational Alerting Rules Matrix

Pipeline Component	Monitored Metric			Threshold Condition	Automated System Action
Streaming Producer	Broker Connectivity			Connection refused	15 retries with exponential back-off; emits critical log.
Speed Layer	Vehicle	Idle	Dwell	$t_{\text{idle}} \geq 60 \text{ seconds}$	Inserts alert record into <code>speed_vehicle_alerts</code> .
Speed Layer	Alert Storming			Duplicate alert < 60 s	Suppresses duplicate alert insertion.
Batch Layer	Daily	Expense	Ingestion	Missing CSV file	Gracefully skips iteration, logs warning without crashing.
Serving API	Database	Query	Latency	DB connection failure	Returns HTTP 500 with detailed diagnostic error response.

7 Demonstration Results & Operational Analytics

7.1 Real-Time Fleet Utilization Findings

The streaming pipeline was validated with 25 concurrent vehicles generating ≈ 750 events per minute. Figure 3 and Table 4 demonstrate the real-time operational metrics across the 4 grid zones.

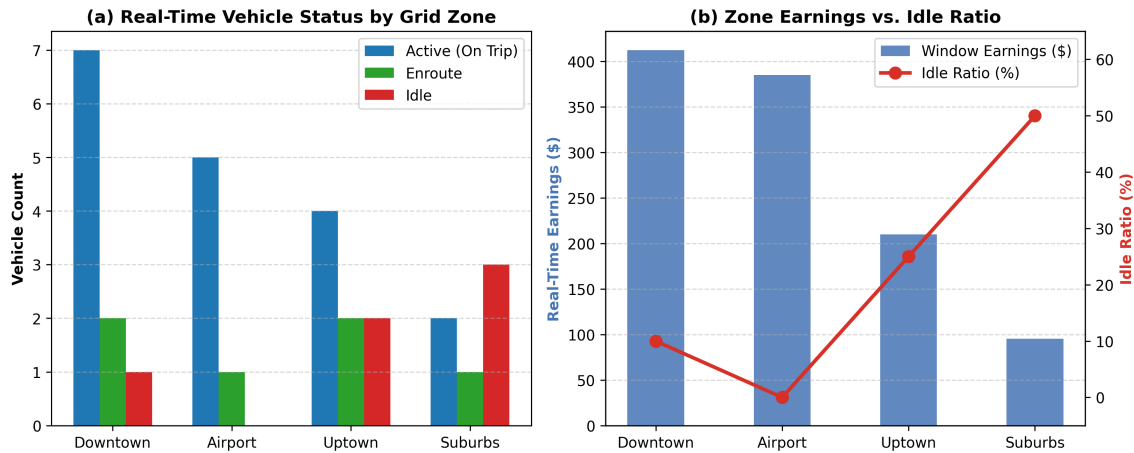


Figure 3: Real-Time Fleet Status Distribution and Zone Earnings vs. Idle Ratios

Table 4: Real-Time Fleet Utilization Sample Output by Grid Zone

Grid Zone	Active	Enroute	Idle	Idle Ratio (%)	Window Revenue (\$)	Avg Speed (km/h)
Downtown	7	2	1	10.0%	\$412.50	32.4
Airport	5	1	0	0.0%	\$385.00	58.2
Uptown	4	2	2	25.0%	\$210.00	28.6
Suburbs	2	1	3	50.0%	\$95.50	44.1

Operational Analysis: The empirical results highlight significant operational asymmetry. While *Downtown* and *Airport* sustain peak fleet efficiency (with idle ratios $\leq 10\%$) and drive the vast majority of fare yields, the *Suburbs* exhibit an alarming 50% idle ratio with subdued earnings. This provides actionable intelligence for dispatch managers to implement dynamic positioning incentives guiding suburban drivers toward urban centers.

7.2 Profitability Reconciliation Analysis

The batch layer reconciliation successfully combined historical fare telemetry with daily garage invoices. Figure 4 and Table 5 illustrate the financial reconciliation across a sample of fleet vehicles.

Business Value Demonstrated: Vehicle **VEH-108** generated respectable gross revenue (\$112.00), but heavy garage maintenance (\$145.00) resulted in a net loss of -\$85.80, triggering an automated recommendation to ground the vehicle for inspection. Vehicle **VEH-114** experienced abnormal fuel burn relative to fare mileage, flagging an immediate engine diagnostic. Without the batch reconciliation layer, these financial losses would have remained completely hidden within gross revenue aggregates.

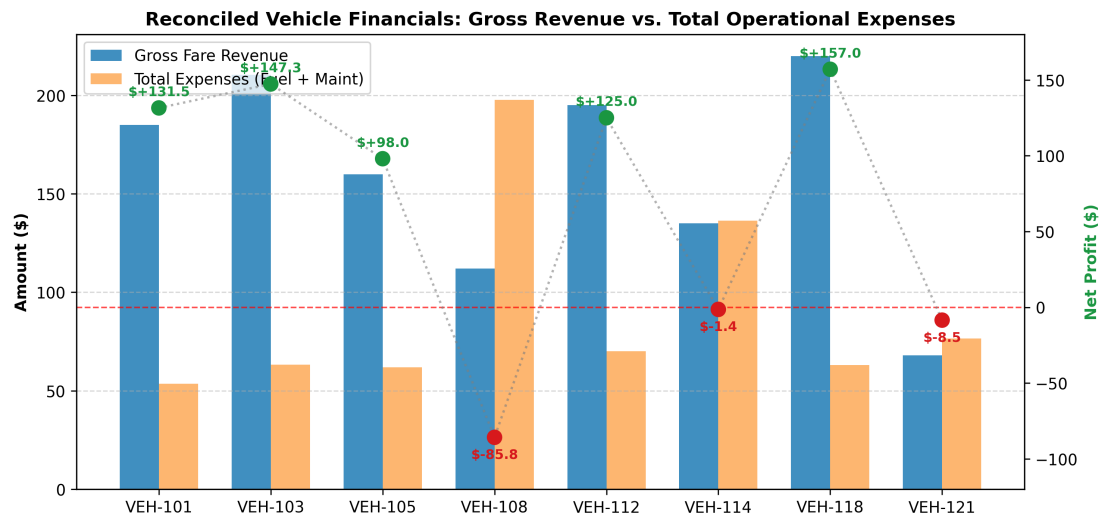


Figure 4: Daily Financial Reconciliation: Gross Fare Revenue vs. Total Operational Expenses

Table 5: Daily Batch Profitability Reconciliation Sample Output

Vehicle ID	Trips	Gross (\$)	Fuel (\$)	Maint (\$)	Net (\$)	Unprofitable?	Operational Prescription
VEH-103	14	\$210.50	\$48.20	\$15.00	+\$147.30	NO	OPTIMAL - High Margin Operations
VEH-108	8	\$112.00	\$52.80	\$145.00	-\$85.80	YES	GROUND VEHICLE - High Maintenance
VEH-114	11	\$135.00	\$118.40	\$18.00	-\$1.40	YES	INSPECT ENGINE - Abnormal Fuel Burn
VEH-121	6	\$68.00	\$41.50	\$35.00	-\$8.50	YES	UNDER-UTILIZED - Relocate Vehicle

8 Limitations, Trade-Offs & Production Scale Roadmap

8.1 Academic Project Constraints & Trade-Offs

While the engineered platform comprehensively fulfills all academic requirements, several trade-offs were consciously made:

1. **Dual Pipeline Maintenance Overhead:** In adhering to classic Lambda design, fare accumulation logic is evaluated in the streaming layer for live estimates and re-executed in the batch layer for historical accuracy. In production, this introduces codebase duplication.
2. **Single-Host Containerization:** All services run via Docker Compose on a single developer workstation. While excellent for reproducibility, it does not demonstrate multi-rack distributed failover.
3. **Clock Compression Artifacts:** Compressing 1 day $\rightarrow$ 5 minutes forces frequent batch DAG triggers, increasing Airflow task scheduling metadata overhead relative to true 24-hour schedules.

8.2 Production Scale Engineering Roadmap

To transition this architecture to enterprise production handling hundreds of thousands of vehicles, the following cloud-native enhancements are recommended:

- **Distributed Processing on Kubernetes (EKS/GKE):** Transition stream processors to Apache Spark Structured Streaming or Apache Flink managed on Kubernetes with dynamic horizontal pod autoscaling.
- **Cloud Object Storage & Open Table Formats (S3 + Apache Iceberg):** Replace local disk directories with AWS S3 and Apache Iceberg/Delta Lake. Iceberg provides ACID transactions, partition evolution, and time-travel querying over petabyte-scale data lakes.
- **Change Data Capture (Debezium):** Ingest transactional changes directly from operational databases via Debezium and Kafka Connect, ensuring zero data loss and automated change event streams.
- **Enterprise Observability (Prometheus & Grafana):** Instrument Kafka JMX metrics, Spark streaming query progress, and database connection pools into Prometheus, providing Grafana dashboards and PagerDuty alert routing.

References

- [1] N. Marz and J. Warren, *Big Data: Principles and Best Practices of Scalable Realtime Data Systems*, Manning Publications, 2015.
- [2] J. Kreps, “Questioning the Lambda Architecture,” *O’Reilly Radar*, 2014. [Online]. Available: <https://www.oreilly.com/radar/questioning-the-lambda-architecture/>
- [3] M. Kleppmann, *Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems*, O’Reilly Media, 2017.
- [4] M. Zaharia, R. S. Xin, P. Wendell, T. Das, M. Armbrust, A. Dave, X. Meng, C. Rosen, S. Venkataraman, M. J. Franklin, A. Ghodsi, J. Gonzalez, S. Shenker, and I. Stoica, “Apache Spark: A Unified Engine for Big Data Processing,” *Communications of the ACM*, vol. 59, no. 11, pp. 56–65, 2016.
- [5] K. Shvachko, H. Kuang, S. Radia, and R. Chansler, “The Hadoop Distributed File System,” in *IEEE 26th Symposium on Mass Storage Systems and Technologies (MSST)*, 2010, pp. 1–10.
- [6] M. Armbrust, T. Das, S. Sun, B. Yavuz, S. Zhu, M. Murthy, J. Torres, H. van Hovell, A. Ionescu, A. Luszczak, M. Switakowski, M. Szafranski, T. Todor, K. Xiao, T. Herman, and M. Zaharia, “Structured Streaming: A Declarative API for Real-Time Applications in Apache Spark,” in *Proceedings of the 2018 International Conference on Management of Data (SIGMOD)*, 2018, pp. 601–613.
- [7] Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna, *EC 8203: Applied Big Data Engineering — Course Syllabus & Mini Project Guidelines*, 2026.